6.13 Expansion Headers

The expansion interface on the board is comprised of two 46 pin connectors. All signals on the expansion headers are **3.3V** unless otherwise indicated.

NOTE: Do not connect 5V logic level signals to these pins or the board will be damaged.

6.13.1 Expansion Header P8

Table 8 shows the default pinout of the **P8** expansion header. Other signals can be connected to this connector based on setting the pin mux on the processor, but this is the default settings on power up. The SW is responsible for setting the default function of each pin.

Table 8. Expansion Header P8 Pinout

SIGNAL NAME	PROC	CONN		PROC	SIGNAL NAME
	GND	1	2	GND	
GPIO1_6	R9	3	4	T9	GPIO1_7
GPIO1_2	R8	5	6	T8	GPIO1_3
TIMER4	R7	7	8	T7	TIMER7
TIMER5	T6	9	10	U6	TIMER6
GPIO1_13	R12	11	12	T12	GPIO1_12
EHRPWM2B	T10	13	14	T11	GPIO0_26
GPIO1_15	U13	15	16	V13	GPIO1_14
GPIO0_27	U12	17	18	V12	GPIO2_1
EHRPWM2A	U10	19	20	V9	GPIO1_31
GPIO1_30	U9	21	22	V8	GPIO1_5
GPIO1_4	U8	23	24	V7	GPIO1_1
GPIO1_0	U7	25	26	V6	GPIO1_29
GPIO2_22	U5	27	28	V5	GPIO2_24
GPIO2_23	R5	29	30	R6	GPIO2_25
UART5_CTSN	V4	31	32	T5	UART5_RTSN
UART4_RTSN	V3	33	34	U4	UART3_RTSN
UART4_CTSN	V2	35	36	U3	UART3_CTSN
UART5_TXD	U1	37	38	U2	UART5_RXD
GPIO2_12	T3	39	40	T4	GPIO2_13
GPIO2_10	T1	41	42	T2	GPIO2_11
GPIO2_8	R3	43	44	R4	GPIO2_9
GPIO2_6	R1	45	46	R2	GPIO2_7

6.13.2 P8 Signal Pin Mux Options

Table 9 shows the other signals that can be connected to each pin of **P8** based on the settings of the registers in the processor for modes 0-3.

Table 9. P8 Mux Options Modes 0-3

PIN	PROC	NAME	MODE0	MODE1	MODE2	MODE3
1		GND				
2		GND				
3	R9	GPIO1_6	gpmc_ad6	mmc1_dat6		
4	T9	GPIO1_7	gpmc_ad7	mmc1_dat7		
5	R8	GPIO1_2	gpmc_ad2	mmc1_dat2		
6	T8	GPIO1_3	gpmc_ad3	mmc1_dat3		
7	R7	TIMER4	gpmc_advn_ale		timer4	
8	T7	TIMER7	gpmc_oen_ren		timer7	
9	T6	TIMER5	gpmc_be0n_cle		timer5	
10	U6	TIMER6	gpmc_wen		timer6	
11	R12	GPIO1_13	gpmc_ad13	lcd_data18	mmc1_dat5	mmc2_dat1
12	T12	GPIO1_12	GPMC_AD12	LCD_DATA19	MMC1_DAT4	MMC2_DAT0
13	T10	EHRPWM2B	gpmc_ad9	lcd_data22	mmc1_dat1	mmc2_dat5
14	T11	GPIO0_26	gpmc_ad10	lcd_data21	mmc1_dat2	mmc2_dat6
15	U13	GPIO1_15	gpmc_ad15	lcd_data16	mmc1_dat7	mmc2_dat3
16	V13	GPIO1_14	gpmc_ad14	lcd_data17	mmc1_dat6	mmc2_dat2
17	U12	GPIO0_27	gpmc_ad11	lcd_data20	mmc1_dat3	mmc2_dat7
18	V12	GPIO2_1	gpmc_clk_mux0	lcd_memory_clk	gpmc_wait1	mmc2_clk
19	U10	EHRPWM2A	gpmc_ad8	lcd_data23	mmc1_dat0	mmc2_dat4
20	V9	GPIO1_31	gpmc_csn2	gpmc_be1n	mmc1_cmd	
21	U9	GPIO1_30	gpmc_csn1	gpmc_clk	mmc1_clk	
22	V8	GPIO1_5	gpmc_ad5	mmc1_dat3		
23	U8	GPIO1_4	gpmc_ad4	mmc1_dat4		
24	V7	GPIO1_1	gpmc_ad1	mmc1_dat1		
25	U7	GPIO1_0	gpmc_ad0	mmc1_dat0		
26	V6	GPIO1_29	gpmc_csn0			
27	U5	GPIO2_22	lcd_vsync	gpmc_a8		
28	V5	GPIO2_24	lcd_pclk	gpmc_a10		
29	R5	GPIO2_23	lcd_hsync	gpmc_a9		
30	R6	GPIO2_25	lcd_ac_bias_en	gpmc_a11		
31	V4	UART5_CTSN	lcd_data14	gpmc_a18	eQEP1_index	mcasp0_axr1
32	T5	UART5_RTSN	lcd_data15	gpmc_a19	eQEP1_strobe	mcasp0_ahclkx
33	V3	UART4_RTSN	lcd_data13	gpmc_a17	eQEP1B_in	mcasp0_fsr
34	U4	UART3_RTSN	lcd_data11	gpmc_a15	ehrpwm1B	mcasp0_ahclkr
35	V2	UART4_CTSN	lcd_data12	gpmc_a16	eQEP1A_in	mcasp0_aclkr

PIN	PROC	NAME	MODE0	MODE1	MODE2	MODE3
36	U3	UART3_CTSN	lcd_data10	gpmc_a14	ehrpwm1A	mcasp0_axr0
37	U1	UART5_TXD	lcd_data8	gpmc_a12	ehrpwm1_tripzone_in	mcasp0_aclkx
38	U2	UART5_RXD	lcd_data9	gpmc_a13	ehrpwm0_synco	mcasp0_fsx
39	T3	GPIO2_12	lcd_data6	gpmc_a6		eQEP2_index
40	T4	GPIO2_13	lcd_data7	gpmc_a7		eQEP2_strobe
41	T1	GPIO2_10	lcd_data4	gpmc_a4		eQEP2A_in
42	T2	GPIO2_11	lcd_data5	gpmc_a5		eQEP2B_in
43	R3	GPIO2_8	lcd_data2	gpmc_a2		ehrpwm2_tripzone_in
44	R4	GPIO2_9	lcd_data3	gpmc_a3		ehrpwm0_synco
45	R1	GPIO2_6	lcd_data0	gpmc_a0		ehrpwm2A
46	R2	GPIO2_7	lcd_data1	gpmc_a1		ehrpwm2B

There are some signals that have not been listed here. Refer to the processor documentation for more information on these pins and detailed descriptions of all of the pins listed. In some cases there may not be enough signals to complete a group of signals that may be required to implement a total interface.

The **PROC** column is the pin number on the processor.
The **PIN** column is the pin number on the expansion header.
The **MODE** columns are the mode setting for each pin. Setting each mode to align with the mode column will give that function on that pin.

Table 10 shows the other **P8** signals for modes 4-7.

Table 10. P8 Mux Options Modes 4-7

PIN	PROC	NAME	MODE4	MODE5	MODE6	MODE7
1		GND				
2		GND				
3	R9	GPIO1_6				gpio1[6]
4	T9	GPIO1_7				gpio1[7]
5	R8	GPIO1_2				gpio1[2]
6	T8	GPIO1_3				gpio1[3]
7	R7	TIMER4				gpio2[2]
8	T7	TIMER7				gpio2[3]
9	T6	TIMER5				gpio2[5]
10	U6	TIMER6				gpio2[4]
11	R12	GPIO1_13	eQEP2B_in			gpio1[13]
12	T12	GPIO1_12	EQEP2A_IN			gpio1[12]
13	T10	EHRPWM2B	ehrpwm2B			gpio0[23]
14	T11	GPIO0_26	ehrpwm2_tripzone_in			gpio0[26]
15	U13	GPIO1_15	eQEP2_strobe			gpio1[15]
16	V13	GPIO1_14	eQEP2_index			gpio1[14]
17	U12	GPIO0_27	ehrpwm0_synco			gpio0[27]
18	V12	GPIO2_1			mcasp0_fsr	gpio2[1]
19	U10	EHRPWM2A	ehrpwm2A			gpio0[22]
20	V9	GPIO1_31				gpio1[31]
21	U9	GPIO1_30				gpio1[30]
22	V8	GPIO1_5				gpio1[5]
23	U8	GPIO1_4				gpio1[4]
24	V7	GPIO1_1				gpio1[1]
25	U7	GPIO1_0				gpio1[0]
26	V6	GPIO1_29				gpio1[29]
27	U5	GPIO2_22				gpio2[22]
28	V5	GPIO2_24				gpio2[24]
29	R5	GPIO2_23				gpio2[23]
30	R6	GPIO2_25				gpio2[25]
31	V4	UART5_CTSN	uart5_rxd		uart5_ctsn	gpio0[10]
32	T5	UART5_RTSN	mcasp0_axr3		uart5_rtsn	gpio0[11]
33	V3	UART4_RTSN	mcasp0_axr3		uart4_rtsn	gpio0[9]
34	U4	UART3_RTSN	mcasp0_axr2		uart3_rtsn	gpio2[17]
35	V2	UART4_CTSN	mcasp0_axr2		uart4_ctsn	gpio0[8]
36	U3	UART3_CTSN			uart3_ctsn	gpio2[16]
37	U1	UART5_TXD	uart5_txd		uart2_ctsn	gpio2[14]

PIN	PROC	NAME	MODE4	MODE5	MODE6	MODE7
38	U2	UART5_RXD	uart5_rxd		uart2_rtsn	gpio2[15]
39	T3	GPIO2_12				gpio2[12]
40	T4	GPIO2_13	pr1_edio_data_out7			gpio2[13]
41	T1	GPIO2_10				gpio2[10]
42	T2	GPIO2_11				gpio2[11]
43	R3	GPIO2_8				gpio2[8]
44	R4	GPIO2_9				gpio2[9]
45	R1	GPIO2_6				gpio2[6]
46	R2	GPIO2_7				gpio2[7]

There are some signals that have not been listed here. Refer to the processor documentation for more information on these pins and detailed descriptions of all of the pins listed. In some cases there may not be enough signals to complete a group of signals that may be required to implement a total interface.

The **PROC** column is the pin number on the processor.
The **PIN** column is the pin number on the expansion header.
The **MODE** columns are the mode setting for each pin. Setting each mode to align with the mode column will give that function on that pin.

6.13.3 Expansion Header P9

Table 11 lists the signals on connector **P9**. Other signals can be connected to this connector based on setting the pin mux on the processor, but this is the default settings on power up. Signals highlighted in yellow are changes from the previous revision of the SRM.

Table 11. Expansion Header P9 Pinout

SIGNAL NAME	PIN	CONN		PIN	SIGNAL NAME
	GND	1	2	GND	
	VDD_3V3EXP	3	4	VDD_3V3EXP	
	VDD_5V	5	6	VDD_5V	
	SYS_5V	7	8	SYS_5V	
PWR_BUT*		9	10	A10	SYS_RESETn
UART4_RXD	T17	11	12	U18	GPIO1_28
UART4_TXD	U17	13	14	U14	EHRPWM1A
GPIO1_16	R13	15	16	T14	EHRPWM1B
I2C1_SCL	A16	17	18	B16	I2C1_SDA
I2C2_SCL	D17	19	20	D18	I2C2_SDA
UART2_TXD	B17	21	22	A17	UART2_RXD
GPIO1_17	V14	23	24	D15	UART1_TXD
GPIO3_21	A14	25	26	D16	UART1_RXD
GPIO3_19	C13	27	28	C12	SPI1_CS0
SPI1_D0	B13	29	30	D12	SPI1_D1
SPI1_SCLK	A13	31	32	VDD_ADC(1.8V)	
AIN4	C8	33	34	GNDA_ADC	
AIN6	A5	35	36	A5	AIN5
AIN2	B7	37	38	A7	AIN3
AIN0	B6	39	40	C7	AIN1
CLKOUT2	D14	41	42	C18	GPIO0_7
	GND	43	44	GND	
	GND	45	46	GND	

*PWR_BUT is a 5V level as pulled up internally by the TPS65217B. It is activated by pulling the signal to GND.

6.13.3.1 Connector P9 Signal Pin Mux Options

Table 12 gives the pin mux options for the signals for connector **P9** for modes 0-3.

Table 12. P9 Mux Options Modes 0-3

PIN	PROC	SIGNAL NAME	MODE0	MODE1	MODE2	MODE3
1		GND				
2		GND				
3		DC_3.3V				
4		DC_3.3V				
5		VDD_5V				
6		VDD_5V				
7		SYS_5V				
8		SYS_5V				
9		PWR_BUT				
10	A10	SYS_RESETn	RESET_OUT			
11	T17	UART4_RXD	gpmc_wait0	mii2_crs	gpmc_csn4	rmii2_crs_dv
12	U18	GPIO1_28	gpmc_be1n	mii2_col	gpmc_csn6	mmc2_dat3
13	U17	UART4_TXD	gpmc_wpn	mii2_rxerr	gpmc_csn5	rmii2_rxerr
14	U14	EHRPWM1A	gpmc_a2	mii2_txd3	rgmii2_td3	mmc2_dat1
15	R13	GPIO1_16	gpmc_a0	gmii2_txen	rmii2_tctl	mii2_txen
16	T14	EHRPWM1B	gpmc_a3	mii2_txd2	rgmii2_td2	mmc2_dat2
17	A16	I2C1_SCL	spi0_cs0	mmc2_sdwp	I2C1_SCL	ehrpwm0_synci
18	B16	I2C1_SDA	spi0_d1	mmc1_sdwp	I2C1_SDA	ehrpwm0_tripzone
19	D17	I2C2_SCL	uart1_rtsn	timer5	dcan0_rx	I2C2_SCL
20	D18	I2C2_SDA	uart1_ctsn	timer6	dcan0_tx	I2C2_SDA
21	B17	UART2_TXD	spi0_d0	uart2_txd	I2C2_SCL	ehrpwm0B
22	A17	UART2_RXD	spi0_sclk	uart2_rxd	I2C2_SDA	ehrpwm0A
23	V14	GPIO1_17	gpmc_a1	gmii2_rxdv	rgmii2_rxdv	mmc2_dat0
24	D15	UART1_TXD	uart1_txd	mmc2_sdwp	dcan1_rx	I2C1_SCL
25	A14	GPIO3_21	mcasp0_ahclkx	eQEP0_strobe	mcasp0_axr3	mcasp1_axr1
26	D16	UART1_RXD	uart1_rxd	mmc1_sdwp	dcan1_tx	I2C1_SDA
27	C13	GPIO3_19	mcasp0_fsr	eQEP0B_in	mcasp0_axr3	mcasp1_fsx
28	C12	SPI1_CS0	mcasp0_ahclkr	ehrpwm0_synci	mcasp0_axr2	spi1_cs0
29	B13	SPI1_D0	mcasp0_fsx	ehrpwm0B		spi1_d0
30	D12	SPI1_D1	mcasp0_axr0	ehrpwm0_tripzone		spi1_d1
31	A13	SPI1_SCLK	mcasp0_aclkx	ehrpwm0A		spi1_sclk
32		VADC				
33	C8	AIN4				
34		AGND				
35	A8	AIN6				

PIN	PROC	SIGNAL NAME	MODE0	MODE1	MODE2	MODE3
36	B8	AIN5				
37	B7	AIN2				
38	A7	AIN3				
39	B6	AIN0				
40	C7	AIN1				
41	D14	CLKOUT2	xdma_event_intr1		tclkin	clkout2
42	C18	GPIO0_7	eCAP0_in_PWM0_out	uart3_txd	spi1_cs1	pr1_ecap0_ecap_capin_apwm_o
43		GND				
44		GND				
45		GND				
46		GND				

There are some signals that have not been listed here. Refer to the processor documentation for more information on these pins and detailed descriptions of all of the pins listed. In some cases there may not be enough signals to complete a group of signals that may be required to implement a total interface.

The **PROC** column is the pin number on the processor.
The **PIN** column is the pin number on the expansion header.
The **MODE** columns are the mode setting for each pin. Setting each mode to align with the mode column will give that function on that pin.

Table 13 gives the pin mux options for the signals for connector **P9** for modes 4-7.

Table 13. P9 Mux Options Modes 4-7

PIN	PROC	SIGNAL NAME	MODE4	MODE5	MODE6	MODE7
1		GND				
2		GND				
3		DC_3.3V				
4		DC_3.3V				
5		VDD_5V				
6		VDD_5V				
7		SYS_5V				
8		SYS_5V				
9		PWR_BUT				
10	A10	SYS_RESETn				
11	T17	UART4_RXD	mmc1_sdcd		uart4_rxd_mux2	gpio0[30]
12	U18	GPIO1_28	gpmc_dir		mcasp0_aclkr_mux3	gpio1[28]
13	U17	UART4_TXD	mmc2_sdcd		uart4_txd_mux2	gpio0[31]
14	U14	EHRPWM1A	gpmc_a18		ehrpwm1A_mux1	gpio1[18]
15	R13	GPIO1_16	gpmc_a16		ehrpwm1_tripzone_input	gpio1[16]
16	T14	EHRPWM1B	gpmc_a19		ehrpwm1B_mux1	gpio1[19]
17	A16	I2C1_SCL				gpio0[5]
18	B16	I2C1_SDA				gpio0[4]
19	D17	I2C2_SCL	spi1_cs1			gpio0[13]
20	D18	I2C2_SDA	spi1_cs0			gpio0[12]
21	B17	UART2_TXD			EMU3_mux1	gpio0[3]
22	A17	UART2_RXD			EMU2_mux1	gpio0[2]
23	V14	GPIO1_17	gpmc_a17		ehrpwm0_synco	gpio1[17]
24	D15	UART1_TXD				gpio0[15]
25	A14	GPIO3_21	EMU4_mux2			gpio3[21]
26	D16	UART1_RXD				gpio0[14]
27	C13	GPIO3_19	EMU2_mux2			gpio3[19]
28	C12	SPI1_CS0	eCAP2_in_PWM2_out			gpio3[17]
29	B13	SPI1_D0	mmc1_sdcd_mux1			gpio3[15]
30	D12	SPI1_D1	mmc2_sdcd_mux1			gpio3[16]
31	A13	SPI1_SCLK	mmc0_sdcd_mux1			gpio3[14]
32		VDD_ADC (1.8V_				
33	C8	AIN4				
34		GNDA_ADC				
35	A5	AIN6				

PIN	PROC	SIGNAL NAME	MODE4	MODE5	MODE6	MODE7
36	A5	AIN5				
37	B7	AIN2				
38	A7	AIN3				
39	B6	AIN0				
40	C7	AIN1				
41	D14	CLKOUT2	timer7_mux1		EMU3_mux0	gpio0[20]
42	C18	GPIO0_7	spi1_sclk	mmc0_sdwp	xdma_event_intr2	gpio0_7
43		GND				
44		GND				
45		GND				
46		GND				

There are some signals that have not been listed here. Refer to the processor documentation for more information on these pins and detailed descriptions of all of the pins listed. In some cases there may not be enough signals to complete a group of signals that may be required to implement a total interface.

The **PROC** column is the pin number on the processor.
The **PIN** column is the pin number on the expansion header.
The **MODE** columns are the mode setting for each pin. Setting each mode to align with the mode column will give that function on that pin.